

Feature- and Score-Space Metrics for Evaluating Deep Speaker Identification: Geometry, Discriminability, Calibration, Ranking Confidence, and Operational Accuracy

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Abstract

Deep speaker-identification systems are often reported with final Top-1 accuracy, equal error rate, or detection cost. These end metrics are necessary but incomplete: they do not explain *why* a speaker embedding succeeds or fails, whether its score space is well separated, whether confidence is meaningful, or whether an apparently strong model is brittle under gallery growth, session mismatch, short duration, or unknown speakers. This paper develops a detailed evaluation framework centered on *feature-based scoring*. We distinguish three levels: (i) embedding geometry, which measures intra-speaker compactness and inter-speaker separation; (ii) score-space discriminability, which evaluates genuine and impostor similarity distributions; and (iii) decision behavior, which converts scores to ranks, identities, or unknown-speaker rejections. We derive and interpret cosine, Euclidean, angular, Mahalanobis, and probabilistic linear discriminant analysis (PLDA) scores; genuine/impostor moments; decidability index d' ; Fisher scatter ratio; within/between-speaker distance ratios; Bhattacharyya distance and distributional overlap; silhouette, Davies–Bouldin, and Calinski–Harabasz indices; Top-1/Top-2 margins and true-speaker margins; likelihood-ratio calibration metrics such as C_{lr} ; score normalization; and score-derived operational measures including ROC/AUC, EER, minDCF, and DIR@FAR. Particular attention is paid to failure modes: non-Gaussian score tails, pair dependence, gallery-size extreme-value effects, speaker imbalance, prototype contamination, and calibration mismatch. We conclude with a reproducible evaluation protocol for deep speaker embeddings and a minimum reporting set that connects representation quality to identification accuracy without conflating the two.

Keywords—speaker identification, speaker embeddings, cosine similarity, PLDA, feature scoring, decidability index, d-prime, Fisher ratio, silhouette score, score margin, calibration, C_{lr} , open-set identification.

1 Introduction

Modern speaker recognition is dominated by neural networks that map a variable-duration utterance to a fixed-dimensional speaker embedding. The x-vector paradigm established discriminatively trained utterance embeddings as a robust alternative to earlier i-vector systems [1]; later speaker-specialized architectures improved temporal aggregation and channel/context modeling [3, 4]. A current systematic review of deep speaker embeddings reports a literature base of hundreds of full-text papers and identifies x-vector, ResNet/ECAPA, Transformer, and hybrid families as major architectural lines [5]. Across these families, the central object at inference is usually not the final training classifier but an embedding $\mathbf{z} \in \mathbb{R}^d$ and a comparison score.

Yet the phrase “speaker-identification accuracy” is often reduced to one number. That is inadequate for engineering diagnosis. Consider two systems with the same Top-1 accuracy. System A may produce genuine scores around 0.85 and nearest-impostor scores around 0.35, whereas System B may produce 0.72 and 0.70. Both can return the same label, but their robustness to noise, gallery growth, or threshold shifts is radically different. Conversely, an embedding can form apparently compact clusters in a two-dimensional visualization while having problematic tail overlap in the actual high-dimensional score space.

This paper therefore treats evaluation as a hierarchy:

$$\begin{aligned} \text{feature geometry} &\rightarrow \text{score discriminability} \\ &\rightarrow \text{decision accuracy} \end{aligned} \quad (1)$$

The first level asks whether embeddings from the same speaker are compact and different speakers are separated. The second asks whether a scoring rule converts those embeddings into well-separated genuine and impostor distributions. The third asks whether a ranking or thresholding policy produces correct identities and rejects unknown speakers at an acceptable false-alarm rate.

The framework is motivated by the history of speaker-evaluation methodology. NIST Speaker Recognition Evaluations formalized threshold-dependent miss/false-alarm costs and DET analysis [8, 9]; Brümmer and du Preez

argued that soft likelihood-ratio outputs require evaluation of both discrimination and calibration [11]; BOSARIS supplied practical tools for DCF, EER, calibration, and large trial lists [12]. Open-set identification benchmarks such as VoxBlink2 now extend the evaluation problem from binary trials to 1: N gallery search with an explicit unknown class [17]. These developments make score-space analysis more, not less, important.

2 Scope and terminology

2.1 Identification, verification, and feature scoring

Let x denote an utterance and f_θ a deep speaker encoder:

$$\mathbf{z} = f_\theta(x), \quad \mathbf{z} \in \mathbb{R}^d. \quad (2)$$

Often the embedding is length normalized:

$$\bar{\mathbf{z}} = \frac{\mathbf{z}}{\|\mathbf{z}\|_2}. \quad (3)$$

In *verification*, two embeddings are compared and a score is thresholded. In *closed-set identification*, a probe is compared to N enrolled speakers and the maximum score wins. In *open-set identification*, the maximum must also exceed a rejection threshold. *Feature-based scoring metrics* are measurements computed directly from embeddings or their similarity/distance scores, before or alongside final identity decisions.

This paper uses the following terms:

- **Genuine score:** similarity between two utterances/prototypes belonging to the same speaker.
- **Impostor score:** similarity between different speakers.
- **Within-speaker variation:** dispersion of embeddings around a speaker center.
- **Between-speaker variation:** separation among speaker centers.
- **Gallery:** enrolled identities and their prototypes/templates.
- **Probe:** utterance to identify.
- **Score calibration:** mapping a raw score to a quantity whose numeric magnitude has a stable probabilistic/decision meaning.

2.2 Why feature metrics are not substitutes for final accuracy

A geometric or distributional metric is a diagnostic. It can correlate strongly with recognition performance, but it does not automatically include the exact gallery size, class priors, rejection threshold, or application costs. A 2024 analysis of clustering-driven self-supervised speaker verification found that clustering measures such as Silhouette and Davies–Bouldin can correlate with downstream EER, illustrating their usefulness while also showing that they are indirect proxies [18]. The correct practice is therefore to report feature/score metrics *and* operational metrics.

3 From deep embeddings to speaker scores

3.1 Enrollment prototypes

For enrolled speaker i with K_i utterances, a common prototype is the normalized mean

$$\mathbf{g}_i = \frac{\sum_{k=1}^{K_i} q_{ik} \bar{\mathbf{z}}_{ik}}{\left\| \sum_{k=1}^{K_i} q_{ik} \bar{\mathbf{z}}_{ik} \right\|_2}, \quad (4)$$

where $q_{ik} \geq 0$ is an optional quality weight. Uniform $q_{ik} = 1$ gives ordinary prototype averaging. Multiple enrollment utterances reduce session-specific variance; open-set benchmarks show clear gains from more enrollment examples [17].

3.2 Cosine similarity

The most common neural-embedding score is

$$s_{\cos}(\mathbf{z}_a, \mathbf{z}_b) = \frac{\mathbf{z}_a^\top \mathbf{z}_b}{\|\mathbf{z}_a\|_2 \|\mathbf{z}_b\|_2}. \quad (5)$$

For length-normalized embeddings, this reduces to an inner product. Deep Speaker explicitly trained embeddings on a hypersphere with cosine-oriented metric learning, illustrating the natural compatibility of angular geometry and speaker identity [2]. Cosine scoring is simple, fast, and well aligned with angular-margin classifier training.

3.3 Angular distance

When only direction matters, define

$$d_{\angle}(a, b) = \arccos(\text{clip}(s_{\cos}, -1, 1)). \quad (6)$$

Angular distance is monotone with cosine similarity, so rank ordering is identical, but the units can be easier to interpret geometrically.

3.4 Euclidean distance

$$d_E(a, b) = \|\mathbf{z}_a - \mathbf{z}_b\|_2. \quad (7)$$

If $\mathbf{z}_a$ and $\mathbf{z}_b$ are unit normalized,

$$d_E^2 = 2 - 2s_{\cos}, \quad (8)$$

so Euclidean and cosine ranking are equivalent. Without normalization, magnitude affects the distance and may encode quality, channel, or nuisance variation.

3.5 Mahalanobis distance

A covariance-aware metric is

$$d_M(a, b) = \sqrt{(\mathbf{z}_a - \mathbf{z}_b)^\top \Sigma^{-1} (\mathbf{z}_a - \mathbf{z}_b)}. \quad (9)$$

It rescales directions according to variability. If estimated from limited data in high dimension, Σ^{-1} is unstable; shrinkage, diagonal covariance, or dimensionality reduction may be required.

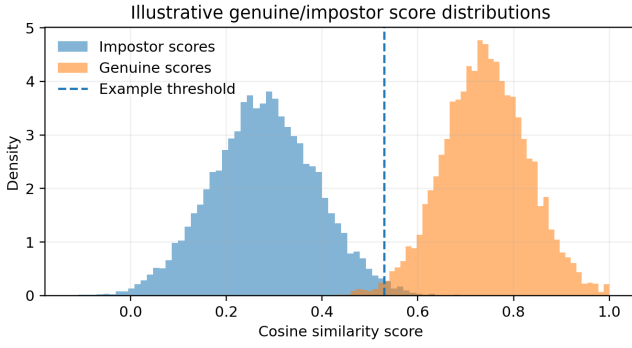


Figure 1: Synthetic illustration of genuine and impostor cosine-score distributions. The plot is conceptual, not a benchmark result. A feature scorer is easier to threshold when separation is large and tail overlap is small.

3.6 PLDA and likelihood-ratio scoring

I-vector systems established PLDA as a strong probabilistic backend, and it remains relevant for learned embeddings [6, 7]. A simplified latent identity model writes

$$\mathbf{z} = \boldsymbol{\mu} + \mathbf{F}\mathbf{y} + \boldsymbol{\epsilon}, \quad (10)$$

where $\mathbf{y}$ is a speaker factor and $\boldsymbol{\epsilon}$ residual/session variation. For two observations, PLDA scoring compares

$$s_{\text{PLDA}} = \log \frac{p(\mathbf{z}_1, \mathbf{z}_2 \mid H_{\text{same}})}{p(\mathbf{z}_1, \mathbf{z}_2 \mid H_{\text{different}})}. \quad (11)$$

A well-calibrated log-likelihood ratio has a direct Bayes-decision interpretation; a raw cosine score does not necessarily have one.

4 Genuine and impostor score distributions

For a scoring function s , construct a genuine set

$$\mathcal{S}_G = \{s(\mathbf{z}_a, \mathbf{z}_b) : y_a = y_b\} \quad (12)$$

and an impostor set

$$\mathcal{S}_I = \{s(\mathbf{z}_a, \mathbf{z}_b) : y_a \neq y_b\}. \quad (13)$$

The central diagnostic question is how much these distributions overlap.

4.1 First- and second-order score statistics

Define

$$\mu_G = \mathbb{E}[S_G], \quad \sigma_G^2 = \text{Var}(S_G), \quad (14)$$

$$\mu_I = \mathbb{E}[S_I], \quad \sigma_I^2 = \text{Var}(S_I). \quad (15)$$

For a similarity score, desirable behavior is high μ_G , low μ_I , and small variances. For a distance score, the direction is reversed. Reporting only means is insufficient: two systems can have the same mean gap but very different tail overlap.

4.2 Decidability index d'

A widely used score-separability measure is

$$d' = \frac{|\mu_G - \mu_I|}{\sqrt{\frac{1}{2}(\sigma_G^2 + \sigma_I^2)}} \quad (16)$$

Higher d' means greater separation in units of pooled standard deviation. The measure is widely used in biometric score analysis and signal detection theory; recent speaker/biometric studies continue to pair it with EER and AUC [24].

4.2.1 Interpretation

If $d' = 0$, the first two moments do not distinguish the distributions. Large d' usually indicates lower overlap. But d' is not an error probability unless additional distributional assumptions hold.

4.2.2 Limitations

- It compresses each distribution to mean and variance.
- Heavy tails or multimodality can make two systems with equal d' have different false-alarm rates.
- Pair scores are not independent when many pairs reuse the same speakers/utterances.
- A global d' can hide speakers with systematically low genuine scores.

Therefore pair d' with empirical tail/threshold metrics.

4.3 Distribution overlap coefficient

A direct overlap measure for estimated densities $p_G(s)$ and $p_I(s)$ is

$$\text{OVL} = \int \min\{p_G(s), p_I(s)\} ds. \quad (17)$$

It lies in $[0, 1]$; lower is better. OVL is intuitively interpretable but density-estimation choices (histogram binning, KDE bandwidth) matter, especially in tails.

4.4 Bhattacharyya coefficient and distance

The Bhattacharyya coefficient is

$$BC = \int \sqrt{p_G(s)p_I(s)} ds, \quad (18)$$

with distance

$$D_B = -\ln BC. \quad (19)$$

For univariate Gaussians, a closed form is

$$D_B = \frac{1}{4} \ln \left[\frac{1}{4} \left(\frac{\sigma_G^2}{\sigma_I^2} + \frac{\sigma_I^2}{\sigma_G^2} + 2 \right) \right] + \frac{1}{4} \frac{(\mu_G - \mu_I)^2}{\sigma_G^2 + \sigma_I^2}. \quad (20)$$

Higher distance means less overlap. Unlike d' , this explicitly responds to unequal variances, though Gaussian assumptions may still be inaccurate.

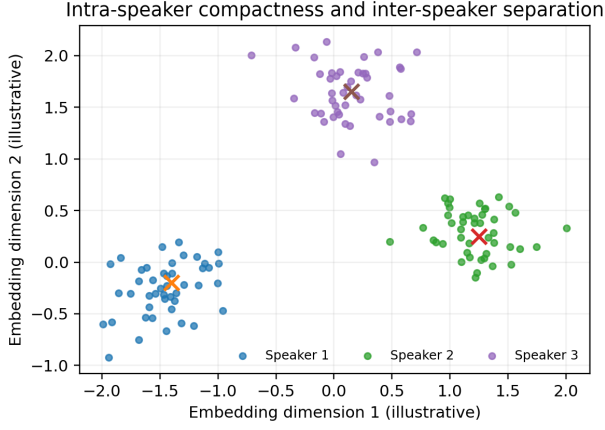


Figure 2: Synthetic 2-D illustration of speaker embedding geometry. Crosses mark speaker centroids. Real embeddings are high dimensional; 2-D plots are qualitative only.

4.5 Jensen–Shannon divergence

With $M = \frac{1}{2}(P_G + P_I)$,

$$D_{JS}(P_G, P_I) = \frac{1}{2}D_{KL}(P_G \| M) + \frac{1}{2}D_{KL}(P_I \| M). \quad (21)$$

This symmetric, finite divergence can quantify score-distribution separation. It is best treated as an analytical diagnostic because its numeric magnitude is less operationally intuitive than FAR/EER.

5 Embedding-space compactness and separation

Score distributions analyze pairs after applying a scorer. Embedding metrics analyze the representation before a specific decision rule.

5.1 Speaker centroids

For speaker i with n_i embeddings,

$$\mathbf{c}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbf{z}_{ij}. \quad (22)$$

For cosine-space analysis, it is often better to normalize samples and then renormalize the centroid.

5.2 Within-speaker scatter

The within-class scatter matrix is

$$S_W = \sum_{i=1}^C \sum_{j=1}^{n_i} (\mathbf{z}_{ij} - \mathbf{c}_i)(\mathbf{z}_{ij} - \mathbf{c}_i)^\top. \quad (23)$$

A scalar compactness index is

$$D_W = \frac{1}{\sum_i n_i} \sum_{i,j} d(\mathbf{z}_{ij}, \mathbf{c}_i). \quad (24)$$

Lower D_W is better, provided the metric d matches deployment scoring.

5.3 Between-speaker scatter

Let global mean $\mathbf{c} = \frac{1}{N} \sum_{i,j} \mathbf{z}_{ij}$. Then

$$S_B = \sum_{i=1}^C n_i (\mathbf{c}_i - \mathbf{c})(\mathbf{c}_i - \mathbf{c})^\top. \quad (25)$$

A simple mean centroid separation is

$$D_B^{(cent)} = \frac{2}{C(C-1)} \sum_{i < k} d(\mathbf{c}_i, \mathbf{c}_k). \quad (26)$$

Higher is better for distance metrics.

5.4 Separation ratio

A useful engineering ratio is

$$R_{sep} = \frac{D_B^{(cent)}}{D_W + \varepsilon}. \quad (27)$$

It is simple and scale dependent, so comparisons should use the same embedding normalization and distance.

5.5 Fisher scatter ratio

A multivariate generalization is

$$J_F = \frac{\text{tr}(S_B)}{\text{tr}(S_W) + \varepsilon}. \quad (28)$$

Higher values indicate more between-speaker energy relative to within-speaker variation. Alternatives use determinants or generalized eigenvalues, but trace ratio is stable and easy to report.

5.6 Speaker-balanced variants

If one speaker has thousands of segments and another has five, pairwise metrics can be dominated by the first. A speaker-balanced within metric is

$$D_W^{bal} = \frac{1}{C} \sum_{i=1}^C \frac{1}{n_i} \sum_{j=1}^{n_i} d(\mathbf{z}_{ij}, \mathbf{c}_i). \quad (29)$$

Speaker-balanced reporting is strongly recommended when class counts vary.

6 Clustering indices as embedding diagnostics

Speaker embeddings naturally form identity clusters. Internal clustering indices can evaluate geometry without using a downstream classifier, while external indices compare clusters to known speaker labels.

6.1 Silhouette coefficient

For point i , let $a(i)$ be its average distance to its own speaker cluster and $b(i)$ its minimum average distance to another cluster. Then

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}. \quad (30)$$

The dataset silhouette is the mean of $s(i)$ and lies in $[-1, 1]$. Higher is better. Rousseeuw introduced the silhouette as a combined cohesion/separation measure [19]. In speaker-recognition research, Silhouette and Davies–Bouldin have been shown to correlate with downstream verification performance in clustering-driven SSL experiments [18].

6.2 Davies–Bouldin index

For cluster i , let S_i be within-cluster scatter and M_{ij} centroid distance. Then

$$R_{ij} = \frac{S_i + S_j}{M_{ij}}, \quad DB = \frac{1}{C} \sum_i \max_{j \neq i} R_{ij}. \quad (31)$$

Lower is better. DB penalizes clusters that are broad relative to their nearest competitor [20].

6.3 Calinski–Harabasz index

A common form is

$$CH = \frac{\text{tr}(S_B)/(C-1)}{\text{tr}(S_W)/(N-C)}. \quad (32)$$

Higher is better [21]. Since it scales with sample size and cluster geometry, it is best for comparing representations on the same evaluation set.

6.4 External clustering metrics

When true speaker labels exist, purity, normalized mutual information (NMI), and adjusted Rand index (ARI) can evaluate cluster-label agreement. These are useful when the identification system contains an explicit clustering stage or when evaluating pseudo-label quality. They are less direct than score metrics for a gallery-based identification system.

6.5 Why t-SNE/UMAP plots are not metrics

Dimensionality-reduction visualizations distort distances and neighborhood structure. They are useful for debugging but should not replace high-dimensional Silhouette, scatter, pair-score, or operational measurements.

7 Score margins for identification confidence

Closed-set identification returns

$$\hat{k} = \arg \max_k s_k, \quad (33)$$

where $s_k = s(\mathbf{z}_p, \mathbf{g}_k)$. Accuracy only records whether $\hat{k}$ is correct. Margin metrics retain more information.

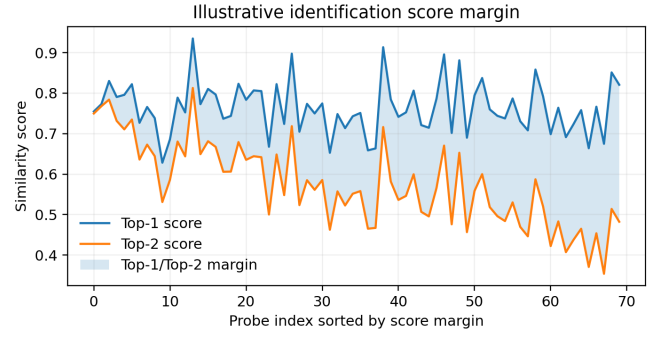


Figure 3: Synthetic illustration of Top-1 and Top-2 gallery scores. The shaded gap is the score margin. Margin is a confidence diagnostic, not a calibrated posterior probability.

7.1 Top-1/Top-2 margin

Let $s_{(1)} \geq s_{(2)}$ be the highest two gallery scores. Define

$$M_{12} = s_{(1)} - s_{(2)}. \quad (34)$$

Large M_{12} means the winner is far ahead of the nearest competitor. Small M_{12} identifies ambiguous probes even when Top-1 is correct.

7.2 Normalized margin

To reduce dependence on raw score scale,

$$M_{norm} = \frac{s_{(1)} - s_{(2)}}{|s_{(1)}| + \varepsilon}. \quad (35)$$

This is convenient but still not invariant to arbitrary non-linear score transformations.

7.3 True-speaker margin

When labels are known during evaluation,

$$M_{true} = s_y - \max_{k \neq y} s_k. \quad (36)$$

This has an especially useful interpretation:

$$M_{true} > 0 \iff \text{correct rank-1 identification}. \quad (37)$$

The distribution of M_{true} gives more insight than accuracy. Values just above zero are fragile correct decisions; strongly negative values are confident errors.

7.4 Per-speaker margin statistics

Report median or lower quantiles of M_{true} by speaker. A global mean can hide a subset of speakers whose prototypes are systematically close to another identity.

8 Prototype quality metrics

Identification often compares a probe to enrollment prototypes rather than individual utterances. Evaluation should therefore measure template quality.

8.1 Enrollment compactness

For speaker i ,

$$Q_i^{comp} = \frac{1}{K_i} \sum_{k=1}^{K_i} s_{\cos}(\tilde{z}_{ik}, \mathbf{g}_i). \quad (38)$$

Higher values indicate internally consistent enrollment. A low value can identify contaminated or heterogeneous enrollment sets.

8.2 Prototype isolation

Define

$$Q_i^{iso} = 1 - \max_{j \neq i} s_{\cos}(\mathbf{g}_i, \mathbf{g}_j). \quad (39)$$

Larger is better. This measures how isolated each enrolled identity is from its nearest enrolled competitor.

8.3 Leave-one-out self-identification

For each enrollment utterance, form a prototype excluding it and score it back to all prototypes. This prevents artificially optimistic self-similarity and is useful before any independent probe set exists.

9 Threshold-free score discrimination metrics

9.1 ROC and AUC

Sweeping threshold τ gives

$$TPR(\tau) = P(S_G \geq \tau), \quad (40)$$

$$FPR(\tau) = P(S_I \geq \tau). \quad (41)$$

AUC is the area under the ROC curve and has the probabilistic interpretation

$$AUC = P(S_G > S_I) \quad (42)$$

for continuous independent samples. It summarizes ranking discrimination over all thresholds but can obscure the very low-FAR region that matters in large galleries.

9.2 DET curves

NIST speaker evaluations popularized DET plots, which display miss and false-alarm probabilities on normal-deviate axes so low-error tradeoffs are visually expanded [8]. DET is not a scalar metric but is often more informative than a single EER.

9.3 Equal error rate

The EER threshold satisfies approximately

$$P_{miss}(\tau) = P_{fa}(\tau). \quad (43)$$

Lower EER indicates stronger pair-score discrimination. It is a useful embedding diagnostic, but it does not model gallery ranking directly. NIST explicitly distinguishes EER from cost-based operating points [10].

10 Decision-cost metrics derived from scores

10.1 Detection cost function

NIST SRE24 defines a normalized cost based on miss and false-alarm probabilities. The unnormalized form is

$$C_{det}(\tau) = C_{miss}P_TP_{miss}(\tau) + C_{fa}(1 - P_T)P_{fa}(\tau), \quad (44)$$

with application parameters (C_{miss}, C_{fa}, P_T) [9]. The minimum over thresholds,

$$\min DCF = \min_{\tau} C_{det}(\tau), \quad (45)$$

measures discriminability at an application-relevant trade-off without requiring the system's submitted threshold to be calibrated.

10.2 Actual DCF versus minDCF

minDCF asks how good the scores could be with the best threshold. *Actual DCF* evaluates a specified threshold and therefore includes calibration. The gap between them is a useful calibration diagnostic.

11 Likelihood-ratio calibration and C_{llr}

Raw similarity is not probability. A score can rank pairs correctly but have numerically unstable thresholds across domains. Calibration maps raw score s to log-likelihood ratio

$$\ell(s) = \log \frac{p(s | H_G)}{p(s | H_I)}. \quad (46)$$

Often a linear logistic map is effective:

$$\ell(s) = as + b. \quad (47)$$

11.1 C_{llr}

Brümmer and du Preez proposed an application-independent logarithmic cost for speaker detectors that output likelihood ratios [11]. One common empirical form in bits is

$$C_{llr} = \frac{1}{2 \ln 2} \left[\frac{1}{N_G} \sum_{i \in G} \ln(1 + e^{-\ell_i}) \right. \quad (48)$$

$$\left. + \frac{1}{N_I} \sum_{j \in I} \ln(1 + e^{\ell_j}) \right]. \quad (49)$$

Lower is better. Perfectly informative calibrated scores approach zero.

11.2 Discrimination and calibration decomposition

Define C_{llr}^{min} as the best monotonic recalibration achievable without changing score ordering. Then

$$C_{llr}^{cal} = C_{llr} - C_{llr}^{min}. \quad (50)$$

$C_{\text{llr}}^{\text{min}}$ reflects discrimination/ranking; $C_{\text{llr}}^{\text{cal}}$ reflects avoidable calibration loss. This decomposition is particularly valuable when two systems have similar EER but different reliability of score magnitude.

11.3 Why calibration matters for identification

For pure closed-set argmax, any strictly monotonic score transform preserves the winner. For open-set identification, however, a global rejection threshold is applied to the top score. Calibration therefore directly affects unknown-speaker decisions and threshold transfer across devices, durations, or galleries.

12 Score normalization

Score normalization is not itself an accuracy metric, but it changes score distributions and therefore the metrics above. Classical speaker recognition uses Z-norm and T-norm to reduce speaker- or test-dependent score variability [13].

12.1 Generic normalized score

$$s' = \frac{s - \mu_c}{\sigma_c + \varepsilon}, \quad (51)$$

where cohort-dependent μ_c, σ_c are estimated from impostor comparisons.

12.2 Z-norm

For each enrolled model, estimate impostor score mean/std using a fixed set of probe cohort utterances. It normalizes model-specific score behavior and can be precomputed.

12.3 T-norm

For each test utterance, score against cohort speaker models and normalize using that test-specific distribution. It adapts to probe condition but costs additional scoring.

12.4 S-norm and adaptive S-norm

Symmetric normalization combines enrollment- and test-side cohort statistics. Adaptive S-norm selects only high-scoring cohort members. Recent work continues to refine score normalization, including embedding-domain alternatives and trainable adaptive normalization [15, 16].

12.5 Evaluation implication

Always state whether reported d' , EER, minDCF, margins, or DIR use raw or normalized scores. Otherwise two results are not reproducible.

13 Feature scoring in open-set identification

Open-set speaker identification (OSSSI) compares a probe to a gallery and must either identify a known speaker or reject it as unknown. VoxBlink2 formalized large-scale OSSSI protocols with multiple gallery sizes and enrollment counts [17].

13.1 Maximum score

$$s^*(p) = \max_{k=1, \dots, N} s(\mathbf{z}_p, \mathbf{g}_k). \quad (52)$$

Decision:

$$\hat{y} = \begin{cases} \arg \max_k s_k, & s^* \geq \tau, \\ \text{unknown}, & s^* < \tau. \end{cases} \quad (53)$$

13.2 DIR at FAR

For known probes K and unknown probes U , a rank- r detection-and-identification rate is

$$\text{DIR}(\tau, r) = \frac{|\{p \in K : \text{rank}_p(y_p) \leq r, s(p, y_p) \geq \tau\}|}{|K|}, \quad (54)$$

while

$$\text{FAR}(\tau) = \frac{|\{p \in U : s^*(p) \geq \tau\}|}{|U|}. \quad (55)$$

Feature-based score quality is operationally summarized by DIR@FAR.

13.3 Extreme-value effect of gallery size

Even if the impostor score distribution for one comparison is unchanged, the maximum of N impostor scores increases with N . Formally,

$$P(\max_{1 \leq k \leq N} S_{I,k} \leq t) = F_I(t)^N \quad (56)$$

under an independence approximation. Therefore

$$P(\max S_I > t) = 1 - F_I(t)^N. \quad (57)$$

This explains why large-gallery identification is much harder than pairwise verification and why a fixed threshold may fail as the gallery grows.

14 Pair construction and statistical dependence

Feature-score metrics are easy to compute incorrectly because pairs are highly dependent.

14.1 Quadratic impostor explosion

If each of C speakers has m utterances, the number of cross-speaker impostor pairs is on the order of $C^2 m^2$, while genuine pairs grow roughly as $C m^2$. A naive average therefore has extreme class imbalance.

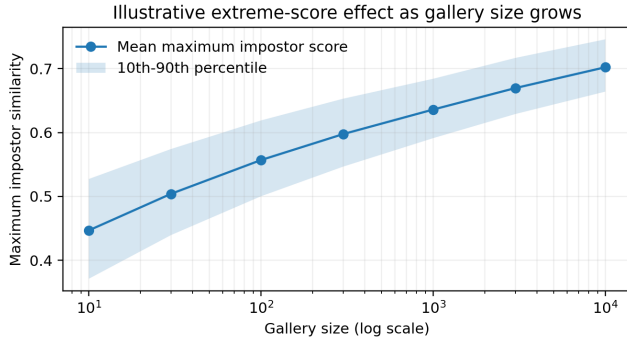


Figure 4: Synthetic extreme-score illustration. Even with an unchanged single-comparison impostor distribution, the maximum impostor score tends to grow with gallery size. Real scores are dependent, so the curve is conceptual.

14.2 Repeated-speaker dependence

Thousands of pair scores can reuse one difficult speaker. Treating them as independent observations produces overconfident confidence intervals. NIST has explicitly studied measurement uncertainty and data dependency in speaker-recognition evaluation and recommends bootstrap-style uncertainty analysis that respects sampling structure [22, 23].

14.3 Recommended resampling unit

For speaker-embedding evaluation, prefer a hierarchical or speaker/session bootstrap:

1. sample speakers with replacement;
2. sample sessions/utterances within those speakers;
3. rebuild genuine/impostor pairs and metrics;
4. repeat to obtain confidence intervals.

This is more defensible than bootstrapping individual pair scores independently.

15 Per-speaker and condition-stratified feature metrics

A single global score distribution can hide systematic weaknesses. Compute metrics by:

- speaker;
- sex/age group if lawful and relevant;
- language/accent;
- microphone/channel;
- SNR/noise condition;
- utterance duration;
- enrollment count;
- session match/mismatch;
- codec or transmission condition.

For speaker i , useful metrics include

$$\mu_{G,i} = \mathbb{E}[s(z_{i,a}, z_{i,b})], \quad (58)$$

$$\mu_{I,i} = \mathbb{E}[s(z_{i,a}, z_{j,b}) \mid j \neq i], \quad (59)$$

$$M_i = \text{median}(M_{\text{true}} \mid y = i). \quad (60)$$

Worst-decile speaker performance is often more operationally meaningful than a global mean.

16 Score-space tail analysis

Large-scale identification failures are driven by tails, not only by means.

16.1 High impostor quantiles

Report upper quantiles such as

$$q_I(0.99), q_I(0.999), q_I(0.9999). \quad (61)$$

A model with slightly lower d' may still be preferable if its extreme impostor tail is much lower.

16.2 Low genuine quantiles

Similarly report

$$q_G(0.01), q_G(0.05). \quad (62)$$

These reveal how often true-speaker similarity collapses under difficult sessions.

16.3 Tail gap

A simple robust separation indicator is

$$G_{\text{tail}} = q_G(0.05) - q_I(0.999). \quad (63)$$

Positive values imply a threshold may exist that accepts at least roughly 95% of genuine scores while rejecting roughly 99.9% of single impostor comparisons, subject to dependence and finite-sample uncertainty. This is a diagnostic, not a standardized benchmark metric.

17 Metric relationships and non-equivalences

Several commonly assumed equivalences are false:

$$\text{high Silhouette} \not\Rightarrow \text{low FAR at a fixed threshold}, \quad (64)$$

$$\text{low EER} \not\Rightarrow \text{high large-gallery DIR}, \quad (65)$$

$$\text{large } d' \not\Rightarrow \text{good calibration}, \quad (66)$$

$$\text{high Top-1 accuracy} \not\Rightarrow \text{large score margin}, \quad (67)$$

$$\text{good visualization} \not\Rightarrow \text{good high-dimensional separation}. \quad (68)$$

Table 1: Feature- and score-space metrics for deep speaker identification. “Direct operational” means the metric directly reflects a decision/ranking policy rather than only representation geometry.

Metric	Space	Direction	What it measures	Key limitation
Mean genuine / impostor score	Score	Gap larger	Basic score separation	Ignores variance and tails
d'	Score	Higher	Mean separation normalized by pooled spread	Moment-based; can miss multimodal/tail overlap
OVL	Score distribution	Lower	Direct density overlap	Sensitive to density estimation
Bhattacharyya distance	Score distribution	Higher	Distribution similarity/separation	Often estimated under simplifying assumptions
AUC	Score/ranking	Higher	Probability genuine outranks impostor	Averages all FPR regions
EER	Score/threshold	Lower	Balanced pairwise verification error	Not 1:N identification accuracy
minDCF	Score/threshold	Lower	Best application-weighted verification cost	Depends on chosen cost/prior
C_{llr}	Calibrated score	Lower	Discrimination plus calibration of LLR scores	Binary-trial framework; requires meaningful LLRs
Within distance D_W	Embedding	Lower	Intra-speaker compactness	Scale/metric dependent
Between centroid distance	Embedding	Higher	Inter-speaker separation	Does not capture local nearest competitors
Fisher ratio	Embedding	Higher	Global between/within scatter	Global average can hide difficult speaker pairs
Silhouette	Embedding clusters	Higher	Cohesion versus nearest cluster separation	Biased by cluster geometry/distance choice
Davies–Bouldin	Embedding clusters	Lower	Worst relative cluster scatter	Not a decision metric
Calinski–Harabasz	Embedding clusters	Higher	Scatter-ratio clustering quality	Sample-size dependent
Top-1/Top-2 margin	Gallery score	Higher	Winner confidence relative to nearest competitor	Not calibrated; scale dependent
True-speaker margin	Gallery score	Higher	Correct-class score relative to best impostor	Requires ground-truth label
Prototype compactness	Enrollment	Higher	Enrollment consistency	Can reward uniformly poor but tight templates
DIR@FAR	Open-set gallery	Higher	Correct known ID at specified unknown false-alarm rate	Protocol/gallery dependent

18 An illustrative diagnostic example

Suppose two encoders achieve 96% closed-set Top-1 accuracy on the same 100-speaker test. Model A has $\mu_G = 0.79$, $\mu_I = 0.27$, $\sigma_G = 0.08$, $\sigma_I = 0.10$. Model B has $\mu_G = 0.73$, $\mu_I = 0.35$, $\sigma_G = 0.11$, $\sigma_I = 0.13$. Then

$$d'_A \approx \frac{0.52}{\sqrt{(0.08^2 + 0.10^2)/2}} \approx 5.74, \quad (70)$$

$$d'_B \approx \frac{0.38}{\sqrt{(0.11^2 + 0.13^2)/2}} \approx 3.15. \quad (71)$$

Equal Top-1 accuracy does not imply equal robustness. Model A has substantially stronger moment-level separation. If Model B nonetheless has better calibration or a lighter extreme impostor tail, the operational conclusion could still differ. Therefore the analysis should continue with score quantiles, EER/minDCF, margins, and open-set DIR@FAR.

19 A recommended feature-score evaluation protocol

19.1 Step 1: freeze the evaluation population

Define:

- speakers and sessions;
- enrollment/probe split;
- closed-set versus open-set condition;
- gallery sizes;
- number of enrollment utterances;
- duration/SNR/channel strata.

Do not tune thresholds or cohorts on the final test set.

19.2 Step 2: extract and store embeddings

Store the raw or normalized embedding with model/version metadata. This permits later rescoring without rerunning the neural encoder.

19.3 Step 3: evaluate embedding geometry

At minimum compute:

1. within-speaker compactness D_W ;
2. between-centroid distance $D_B^{(cent)}$;
3. Fisher ratio J_F ;
4. Silhouette and Davies–Bouldin;
5. per-speaker distributions of the above.

19.4 Step 4: build genuine and impostor score sets

Use the exact deployment scorer (cosine, PLDA, or other). Construct speaker-balanced summaries to prevent a few prolific speakers dominating.

19.5 Step 5: score-space diagnostics

Report:

- $\mu_G, \sigma_G, \mu_I, \sigma_I$;
- d' ;
- genuine/impostor histograms or KDE;
- high impostor and low genuine quantiles;
- optional OVL/Bhattacharyya distance;
- ROC/DET and EER;
- minDCF at declared priors/costs.

19.6 Step 6: gallery margin diagnostics

For every probe compute $s_{(1)}, s_{(2)}, M_{12}, M_{true}$. Report median and 5th percentile margins, not only the mean.

19.7 Step 7: calibration

If scores are intended as likelihood ratios or thresholds must transfer across conditions, calibrate on development data and report C_{llr} , C_{llr}^{min} , and calibration loss. State whether score normalization is applied.

19.8 Step 8: open-set evaluation

Use known and unknown probe speakers disjoint from training as appropriate. Report DIR@FAR for several gallery sizes and enrollment counts. VoxBlink2 provides a modern example of such protocol design [17].

19.9 Step 9: uncertainty

Use speaker/session-aware bootstrap confidence intervals. For system comparisons, report paired bootstrap differences because the same probes are evaluated by both systems.

20 Minimum reporting set

For a research paper or engineering acceptance test centered on feature-based scoring, the following set is compact but informative:

1. **Scorer definition:** cosine/PLDA/etc., embedding normalization, score normalization.
2. **Genuine/impostor moments:** $\mu_G, \sigma_G, \mu_I, \sigma_I$.
3. **Score separability:** d' and a score-distribution plot.
4. **Embedding geometry:** $D_W, D_B^{(cent)}$, Fisher ratio, Silhouette.
5. **Tail statistics:** low genuine and high impostor quantiles.
6. **Ranking confidence:** true-speaker margin and Top-1/Top-2 margin.
7. **Pairwise operational diagnostic:** EER and minDCF.
8. **Calibration:** C_{llr} or actual-vs-minDCF gap when thresholds matter.
9. **Identification:** Top-1/Top- K for closed set; DIR@FAR for open set.
10. **Stratification/uncertainty:** per-speaker/condition breakdown plus confidence intervals.

This set makes it possible to distinguish a representation problem from a scoring problem, a calibration problem, and a gallery/search problem.

21 Implementation details and numerical stability

21.1 Use the deployment normalization

If inference length-normalizes embeddings, evaluate feature metrics on the normalized representation as well. A model can look different before and after projection to the unit hypersphere.

21.2 Avoid all-pairs memory explosion

Do not materialize a $M \times M$ matrix for very large corpora unless necessary. Compute impostor statistics in blocks or sample speaker-balanced impostor pairs. For tail evaluation, sampling must be sufficiently large to estimate the target quantile.

21.3 Stable variance and covariance

Compute variance with numerically stable algorithms. For Mahalanobis/Fisher metrics in high dimension, regularize covariance:

$$\Sigma_\lambda = (1 - \lambda)\hat{\Sigma} + \lambda \frac{\text{tr}(\hat{\Sigma})}{d} I. \quad (72)$$

21.4 Distance consistency

Do not compute Silhouette with Euclidean distance while deployment uses cosine unless there is a clear reason. On unit vectors Euclidean and cosine are monotonic, but with unnormalized embeddings they need not agree.

21.5 Confidence intervals for d'

Because pair dependence invalidates simple iid formulas, prefer hierarchical bootstrap. Report, for example,

$$d' = 4.21 \quad [95\% \text{ CI} : 3.98, 4.45]. \quad (73)$$

The confidence interval is often more important than the second decimal place.

22 Common failure modes

22.1 Using the test set to choose thresholds

This leaks evaluation information and produces optimistic FAR/DIR. Thresholds and calibration parameters belong to development data.

22.2 Reporting only a global d'

A high global d' can coexist with a subset of confusable speakers. Add nearest-centroid separation and per-speaker margin distributions.

22.3 Assuming score Gaussianity

d' and Gaussian Bhattacharyya formulas use low-order summaries. Inspect empirical histograms, quantiles, and DET tails.

22.4 Ignoring score direction

Some backends output similarity (higher is same speaker); others output distance (lower is same speaker). Normalize conventions before computing margins or genuine/impostor gaps.

22.5 Mixing raw and normalized scores

Score normalization can substantially change overlap and tails [13, 14]. Always label the stage at which a metric is computed.

22.6 Using one threshold across changing gallery sizes

Open-set maximum-impostor behavior shifts with gallery size (Fig. 4). Recalibration or gallery-aware threshold analysis may be required.

22.7 Prototype contamination

An enrollment set containing another speaker can produce a deceptively central but wrong prototype. Enrollment compactness and leave-one-out checks should precede deployment.

22.8 Confusing calibration with discrimination

A monotonic recalibration can improve C_{1lr} or actual DCF without changing rank, AUC, EER, or Top-1. This is not contradictory: calibration and discrimination are different properties [11, 12].

23 Relation to deep-network training objectives

Feature metrics can reveal what training losses are doing geometrically.

23.1 Softmax and angular-margin objectives

Speaker-classification training encourages between-class separation. Angular-margin objectives further enforce hyperspherical margins, often improving cosine-score separation. Their success should be visible as higher genuine/impostor d' , higher M_{true} , and improved tail gap, not merely lower training loss.

23.2 Metric learning

Triplet/contrastive objectives explicitly act on pair or tuple geometry. Deep Speaker showed a cosine-hypersphere formulation where the learned embedding is directly used for identification, verification, and clustering [2]. Evaluation should therefore include pair-score and cluster metrics aligned with the loss.

23.3 Attentive pooling and within-speaker variance

Attentive statistics pooling weights informative frames and models weighted standard deviation, improving the discriminative quality of utterance embeddings [3]. One expected diagnostic effect is reduced within-speaker scatter under variable phonetic content, though this must be measured rather than assumed.

23.4 Robust architectures and condition-specific metrics

Architectural gains such as ECAPA-TDNN's channel attention and multi-scale aggregation are intended to improve robust speaker representations [4]. Condition-stratified D_W , d' , and margin statistics can show whether robustness gains come from stabilizing genuine scores, suppressing impostor scores, or both.

24 Feature-scoring metrics for model selection

A practical model-selection objective can combine standardized validation metrics rather than optimize only EER. For example,

$$Q = w_1 \widetilde{d'} + w_2 \widetilde{J_F} + w_3 \widetilde{M_{true}} - w_4 \widetilde{EER} - w_5 \widetilde{\min DCF}, \quad (74)$$

where tildes denote normalized metrics across candidate checkpoints. This is not a universal standard; it is an engineering technique to prevent selecting a checkpoint that improves one aggregate number while degrading margin or cluster stability. The final model should still be chosen against the declared operational objective.

25 Reproducibility checklist

Every feature-score evaluation should freeze the following:

- encoder checkpoint hash and embedding layer;
- embedding dimension and L2-normalization policy;
- VAD, feature extraction, and utterance duration rules;
- enrollment prototype construction;
- scorer and any LDA/PLDA transform;
- cohort set and Z/T/S/AS normalization configuration;
- pair-generation and speaker-balancing rules;
- unknown-speaker population for open-set tests;
- gallery sizes and number of enrollment utterances;
- threshold/calibration development set;
- random seeds for pair sampling/bootstrap;
- confidence-interval method.

Without these details, feature-score metrics are difficult to reproduce and easy to compare incorrectly.

26 Discussion

Feature-based scoring metrics are valuable because they reveal failure mechanisms hidden by final accuracy. A system can fail because embeddings from the same speaker are unstable (D_W too high), speaker centers are too close (D_B too low), a backend transforms geometry poorly (low d' despite reasonable clusters), scores are discriminative but miscalibrated (C_{llr}^{cal} high), or a large gallery creates extreme impostor maxima even though pairwise EER is low. These are different engineering problems and require different interventions.

No single feature metric should become a new “accuracy” number. d' is compact but moment-based. Silhouette is useful but cluster-shaped. Fisher ratio is global. Score margins are directly tied to identification but uncalibrated. EER is widely understood but binary. C_{llr} is principled for likelihood-ratio outputs but not a direct multi-class identification rate. DIR@FAR is operationally compelling

for open-set SID but protocol dependent. The correct strategy is a layered metric panel.

The strongest conceptual distinction is between *separation* and *calibration*. A scoring system can preserve perfect ordering yet have a useless numeric scale; conversely, a perfectly calibrated system with poor separation still cannot identify reliably. Modern speaker-recognition evaluation should therefore treat embeddings, scores, and decisions as separately measurable objects.

27 Conclusion

Feature-based scoring provides a detailed lens on speaker-identification accuracy that is unavailable from Top-1 alone. The central quantities are intra-speaker compactness, inter-speaker separation, genuine/impostor score overlap, tail behavior, gallery margins, and calibration. Cosine or PLDA scores should be analyzed through empirical score distributions; d' offers a compact separation statistic; Fisher and cluster indices measure embedding geometry; true-speaker and Top-1/Top-2 margins diagnose identification ambiguity; C_{llr} separates discrimination from calibration for likelihood-ratio systems; and DIR@FAR connects feature scores to open-set operation.

A rigorous evaluation therefore follows the chain

$$\begin{aligned} \text{embedding geometry} &\rightarrow \text{pair-score separation} \\ &\rightarrow \text{gallery margins} \rightarrow \text{calibration.} \\ &\rightarrow \text{operational identification} \end{aligned} \quad (75)$$

When these layers are reported together, researchers can determine not only whether a speaker-identification system is accurate, but also whether its learned representation is intrinsically discriminative, whether its scores are trustworthy, and how robust the system is likely to remain when the deployment population and acoustic conditions change.

A Formula reference

A.1 d'

$$d' = \frac{|\mu_G - \mu_I|}{\sqrt{(\sigma_G^2 + \sigma_I^2)/2}}.$$

A.2 Within/between separation ratio

$$R_{sep} = \frac{D_B^{(cent)}}{D_W + \varepsilon}.$$

A.3 Fisher trace ratio

$$J_F = \frac{\text{tr}(S_B)}{\text{tr}(S_W) + \varepsilon}.$$

A.4 Silhouette

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}.$$

A.5 Top-1/Top-2 margin

$$M_{12} = s_{(1)} - s_{(2)}.$$

A.6 True-speaker margin

$$M_{true} = s_y - \max_{k \neq y} s_k.$$

A.7 Open-set maximum score

$$s^* = \max_k s(\mathbf{z}_p, \mathbf{g}_k).$$

A.8 Detection cost

$$C_{det} = C_{miss}P_T P_{miss} + C_{fa}(1 - P_T)P_{fa}.$$

B Example machine-readable result schema

A feature-score evaluation result can be stored as:

```
{
  "embedding": {
    "dim": 192,
    "l2_normalized": true,
    "within_distance": 0.31,
    "between_centroid_distance": 0.88,
    "fisher_trace_ratio": 7.4,
    "silhouette": 0.46
  },
  "score": {
    "backend": "cosine",
    "mu_genuine": 0.79,
    "sigma_genuine": 0.08,
    "mu_impostor": 0.27,
    "sigma_impostor": 0.10,
    "d_prime": 5.74,
    "eer": 0.012,
    "min_dcf": 0.074
  },
  "identification": {
    "median_true_margin": 0.21,
    "top1_accuracy": 0.96,
    "gallery_size": 100
  }
}
```

All numeric values above are illustrative only.

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